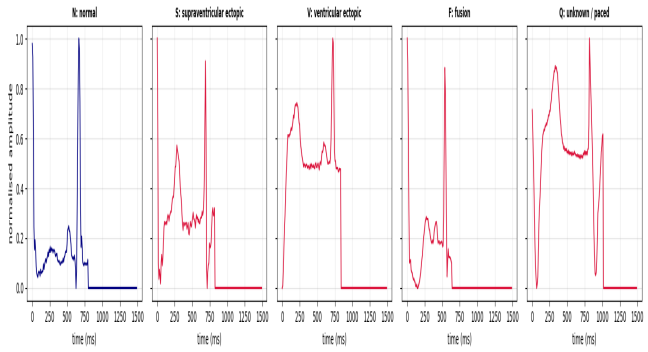
CLASSIFYING HEARTBEATS FROM ECG SIGNALS

A Multilayer Perceptron Built & Trained from First Mathematical Principles in PyTorch

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1. CLINICAL PROBLEM & DATASET

Cardiac arrhythmias are crucial indicators of cardiac dysfunction. Using the MIT-BIH Arrhythmia benchmark (125 Hz, 187 samples/beat), we classify single-lead heartbeats across five AAMI standard categories: Normal Sinus (N), Supraventricular (S), Ventricular (V), Fusion (F), and Paced/Unknown (Q).



Severe 113:1 Class Imbalance:

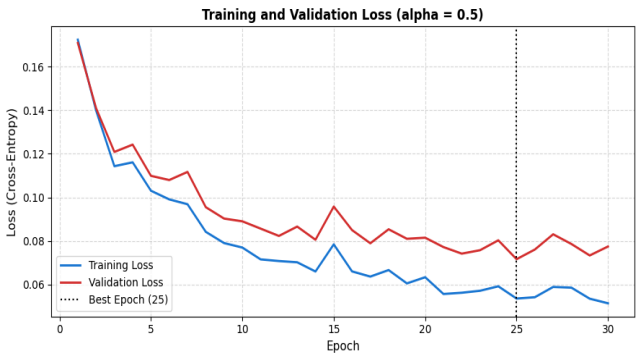
Class	Category	Train	Share
0 (N)	Normal Sinus	72,471	82.8%
1 (S)	Supraventricular	2,223	2.5%
2 (V)	Ventricular Ectopic	5,788	6.6%
3 (F)	Fusion Beat	641	0.7%
4 (Q)	Paced / Unknown	6,431	7.4%

2. FIRST-PRINCIPLES IMPLEMENTATION

The model was implemented strictly from scratch without nn.Linear, nn.Sequential, or torch.optim.

Topology: 187 → 128 → 64 → 5 (32,645 weights)
He Init: W ~ N(0, 2/n_in), b = 0
Stable Loss: L = log(∑ exp(z_j - max(z))) + max(z) - z_y
Manual SGD: v_t = βv_{t-1} + g_t + λθ_t, θ_{t+1} = θ_t - αv_t

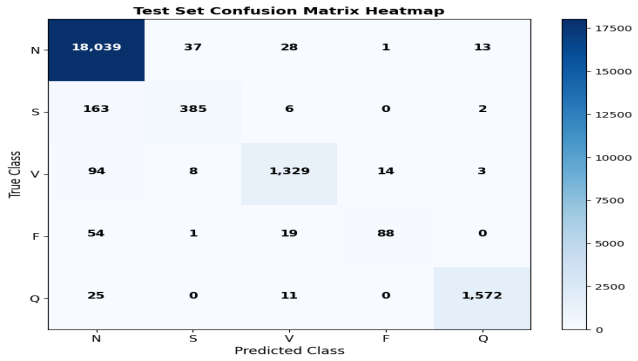
3. TRAINING DYNAMICS & LR SWEEP



Learning rate α = 0.01 achieved optimal monotonic convergence (val loss 0.0924). In contrast, α = 0.1 exhibited gradient oscillations, while α = 0.001 underfit.

4. EMPIRICAL EVALUATION

Test Accuracy: 97.81% | Balanced Accuracy: 82.53%



Class	Precision	Recall	F1-Score
0 (N)	98.4%	99.5%	98.9%
1 (S)	78.4%	61.2%	68.7%
2 (V)	94.6%	93.3%	94.0%
3 (F)	68.2%	54.3%	60.5%
4 (Q)	99.0%	97.5%	98.2%

5. Key Findings: Normal and Paced beats exhibit distinctive morphological signatures (>98% F1). Rare Fusion and Supraventricular beats exhibit lower recall (54%–61%) due to physiological overlap with sinus waveforms.